1 Learning objectives

In this experiment you will investigate:

- the influence of saturation on inductive current transformers
- the influence of the burden on the measuring accuracy

2 References

[1] IEC 61869-2 „Instrument transformers - Part 2: Additional requirements for current transformers“, 2012-09

3 Introduction

The transfer function of a current transformer (CT, see Figure 1) is mainly characterised by the standardised errors: composite error ε_c, current error ε_I and phase displacement δ_I. For the definition of the standardised errors, see the lecture notes.

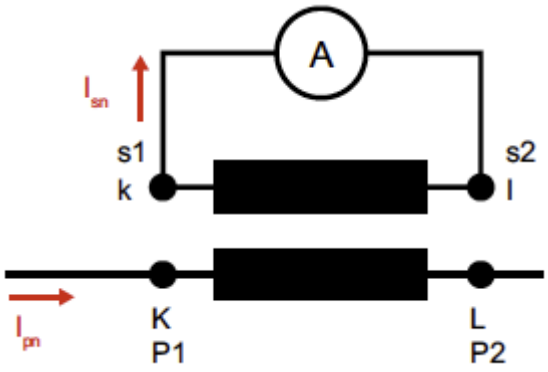


Figure 1: Equivalent circuit diagram of an inductive current transformer.

Inductive CTs with an iron core go into saturation above a certain current level, depending on the loading. Due to saturation, the secondary currents can be very strongly distorted and influence the operation of the measuring and protective devices. For example, currents that are too high cannot be detected correctly.

The burden (Z_B) is the total impedance of the secondary circuit, i.e. the impedance of the protection or measuring devices as well as that of cabling and possibly an additional burden resistor. It is usually expressed in Ω or VA with an indication of the power factor (typically in the range 0.8 ind. to 1).

The overcurrent behaviour of the core is characterised by two factors:

1. instrument security factor (FS) for measuring transformers
2. accuracy limit factor (n_N) for protective CTs.

These values are specified on the rating plate for a load with the secondary rated burden Z_{BN}. If the actual secondary burden of the CT deviates from the rated burden, the actual accuracy limiting factor or the overcurrent limiting factor will also change.

The secondary limiting e.m.f. U_H is the product of the accuracy limit factor n_N , the secondary rated current I_{2N} and the geometric sum of the rated burden Z_{BN} and the internal impedance of the secondary winding ($Z_i \approx R_i$).

In addition to current and angle errors, saturated CTs exhibit large waveform errors in the secondary current.

4 Test program

4.1.1 Device Under Test (DUT)

The following inductive CT is used as DUT:

Before carrying out the test, measure the internal impedance R_x of the CT at room temperature using a Thomson bridge. Figure 2 shows the connection diagram for the measurement with the Thomson bridge. The current transformer must be isolated from the test setup. Also measure the available burden resistances including the cabling.

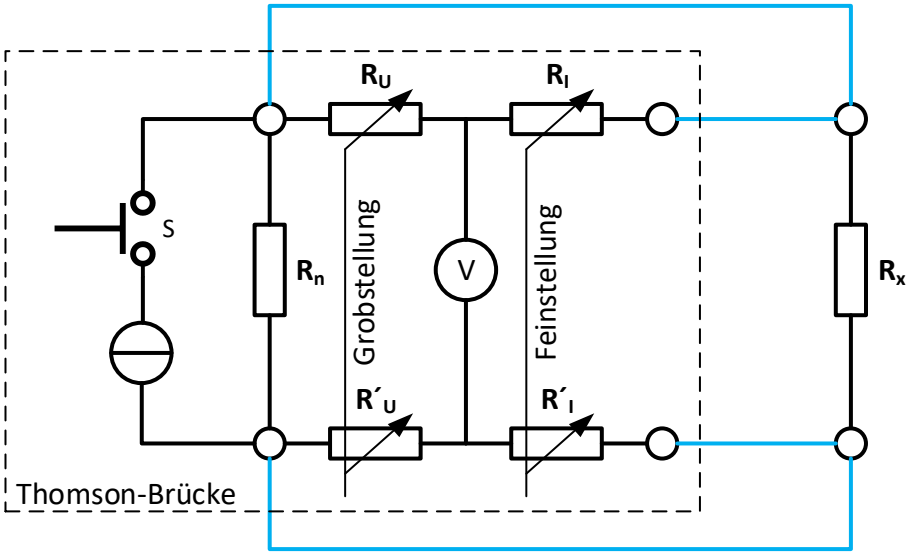


Figure 2: Connection of the analogue Thomson bridge to determine an unknown impedance. Rx is the connected device under test.

Table 1: Internal resistance R_i of the device under test and impedance of the available burdens including cabling.

R_i seen from primary	R_i seen from secondary	Z_{B1}	Z_{B2}	Z_{B3}
..... Ω	..... Ω	..... Ω	..... Ω	..... Ω

4.2 Direct measurement

4.2.1 Test setup

In accordance with IEC 61869-2, the total measurement deviation can be measured directly using the following circuit (see Figure 2). Only TRMS (true RMS) measuring devices should be used for current and voltage measurement. To measure current with the oscilloscope, use current clamps that you have calibrated beforehand. The current clamps (labelled Ch1 and Ch2 in Figure 3) can be used to measure the phase angle between the primary and secondary current.

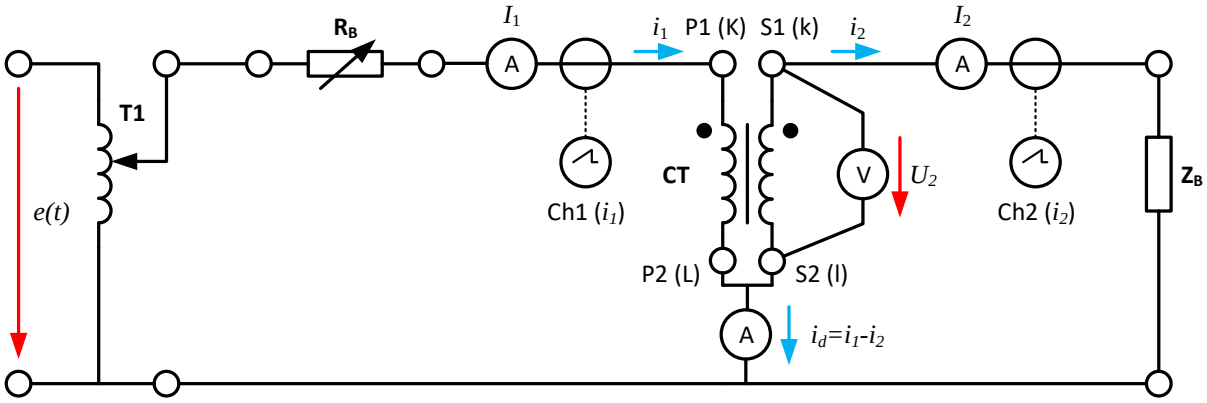


Figure 3: Basic circuit for direct measurement of the total measurement deviation for current transformers with transformation ratio 1. T1 - variable transformer, R_B - current limiting resistor (10Ω, 13A), CT - test object, current transformer, Z_B - burden

4.2.2 Measurement with rated burden

Connect the burden $Z_B \approx Z_{BN} =$ ..... Ω an. Measure the primary currentI_1, the secondary current I_2 and the residual current I_d as well as the secondary voltage U_2. Read the time shift Δt between the primary and secondary current from the oscilloscope. Enter your measurements in Table 2. The shaded columns are intended for post-processing the measurement results.

4.2.3 Measurement with overloading

Now connect the load Bürde $Z_B > Z_{BN}, Z_B =$ ..... Ω. Measure the primary, secondary and differential current as well as the secondary voltage. Read the time shift between the primary and secondary current from the oscilloscope. Enter your measurements in Table 3.

4.2.4 Measurement with underloading

Now connect the burden $Z_B < Z_{BN}, Z_B =$ ..... Ω. Measure the primary, secondary and differential current as well as the secondary voltage. Read the time shift between the primary and secondary current from the oscilloscope. Enter your measurements in Table 4.

4.3 Indirect measurement

4.3.1 Test setup

According to IEC 61869-2, the total measurement deviation can be indirectly approximated using the following circuit (Figure 4). Only TRMS (true RMS) measuring devices should be used for current and voltage measurement. Also record the secondary voltage with the oscilloscope.

With the primary winding open, the secondary winding is connected to a practically sinusoidal voltage with a rated frequency of 50 Hz (see Figure 4). The supply voltage should be gradually increased until the secondary limit e.m.f. U_H has been reached or the current I_2 has reached approx. 20% of the secondary rated current.

What value do you expect for the secondary limit e.m.f. U_H? Select the tap position of the high-current transformer T2 accordingly.

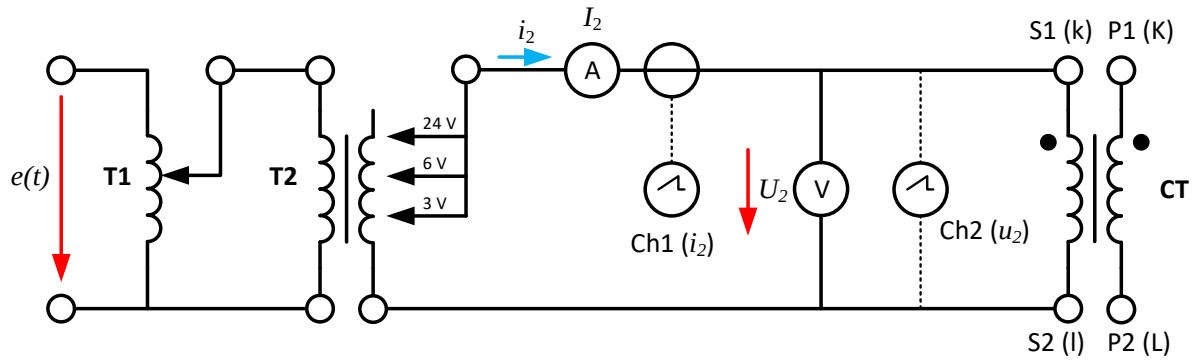


Figure 4: Basic circuit for indirect measurement of the total composite error.
T1 - variable transformer, T2 - high-current transformer, CT - test object

4.3.2 Measurements

The characteristic curve $I_2 = f(U_2)$ must first be recorded and entered in Table 5. Enter your measurements there. The shaded columns are intended for post-processing the measurement results.

Note: The measured values are not evenly distributed. In order to obtain more accurate measured values at the saturation range of the characteristic curve, you should measure more densely in the saturation range, i.e. with smaller current steps.

5 Post-processing

5.1 Direct measurement

a) Calculate from the measured values in Table 2 the current error ε_I, the phace displacement δ_I and the composite error ε_C for operation with rated burden.
The measurement errors according to IEC 61869-2 are listed in the following equations.

$$\varepsilon_I = \frac{\Delta I}{I_1} = \frac{K_n \cdot I_2 - I_1}{I_1}$$

$$\delta_I = \arg\left(\frac{I_2}{I_1}\right) = \arg(I_2) - \arg(I_1)$$

$$\varepsilon_C = \frac{I_d}{I_1} = \frac{1}{I_1}\sqrt{\frac{1}{T}\int_0^T \left(K_n \cdot i_2(t) - i_1(t)\right)^2 dt}$$

b) Enter the current measurement error $\varepsilon_I = f(I_1)$ and the phase displacement $\delta_I = f(I_1)$ as a function of primary current up to $120\ \%\ I_{1n}$ at rated burden in one diagram each. Also enter the required values for a measuring transformer of accuracy class 1. Are the requirements met?

c) Calculate the above measurement errors for the case of overload and underload and enter them in Table 3 and Table 4.

d) Plot the measured secondary current $I_2 = f(I_1)$ over the primary current at different burden in a diagram. What relationship can you recognise between load and measurement error?

e) Plot the composite error $\varepsilon_C = f(I_1)$ over the primary current at different loads in a diagram. Determine the rated overcurrent limiting factor FS from the characteristic curve for rated burden.

f) Calculate the secondary-side limit voltage U_H from the overcurrent limiting FS.

g) Determine the overcurrent limiting factor FS_x from the diagram in e) for different burdens. Compare the factors determined in this way with those calculated directly from the formula in the lecture notes for the different burdens.

h) Plot the composite error $\varepsilon_C = f(U_2)$ across the secondary voltage with different loads in another diagram. What can you learn from this diagram?

5.2 Indirect measurement

i) From the measurements in Table 5, plot the secondary voltage $U_2 = f(I_2)$ as a function of secondary current in a diagram. Determine the secondary-side saturation voltage U_L from the diagram. Compare this value with the value of the limit voltage U_H from question f).

j) Use Table 5 to calculate the expected secondary current I_2' and the expected composite error ε_C for rated burden or for over- or under-loading according to the following equations. Enter the calculated values in Table 5.

$$I_2' = \frac{U_2}{Z_B + R_i}$$

$$\varepsilon_C = \frac{I_2}{I_2'}$$

k) Plot the calculated total measurement deviation $\varepsilon_C = f(I_2')$ over the expected secondary current. Determine the overcurrent limiting factor FS for all three values of the burden.

l) Compare the values determined under e), g) and k) for the overcurrent limiting factors. Explain any deviations between the measured and calculated values. Compare the indirect with the direct measurement method.

Annex

Table 2: Direct measurement with rated burden.

Table 3: Direct measurement with overloading

Table 4: Direct measurement with underloading

Table 5: Secondary current I_2 in A, secondary voltage U_2 in V, calculated secondary current at load I_2' in A, calculated composite error ε_C in %.

Table 2: Direct measurement with rated burden.

primary current	I_1 in A	I_2 in A	ΔI in A [1]	ε_I in %	I_d in A	ε_C in %	U_2 in V	Δt in µs	δ_I in Minuten
5 % I_{1n}									
20 % I_{1n}									
40 % I_{1n}									
60 % I_{1n}									
80 % I_{1n}									
100 % I_{1n}									
120 % I_{1n}									
max. 10A									

[1] $\Delta I = I_1 - I_2$

Table 3: Direct measurement with overloading.

primary current	I_1 in A	I_2 in A	ΔI in A	ε_I in %	I_d in A	ε_C in %	U_2 in V	Δt in µs	δ_I in Minuten
5 % I_{1n}									
20 % I_{1n}									
40 % I_{1n}									
60 % I_{1n}									
80 % I_{1n}									
100 % I_{1n}									
120 % I_{1n}									
max. 10A									

Table 4: Direct measurement with underloading.

primary current	I_1 in A	I_2 in A	ΔI in A	ε_I in %	I_d in A	ε_C in %	U_2 in V	Δt in µs	δ_I in Minuten
5 % I_{1n}									
20 % I_{1n}									
40 % I_{1n}									
60 % I_{1n}									
80 % I_{1n}									
100 % I_{1n}									
120 % I_{1n}									
max. 10A									

Table 5: Secondary current I_2 in A, secondary voltage U_2 in V, calculated secondary current at load I_2' in A, calculated composite error ε_C in %.

I_2 in A	U_2 in V	Rated burden		Overloading		Underloading	
		I_2' in A	ε_C in %	I_2' in A	ε_C in %	I_2' in A	ε_C in %
up to 20% of I_{2n}							